SOLVE FOR A TOTAL OF 50 POINTS. SHOW COMPLETE SOLUTIONS.

1. FIXEDPOINT THEOREM (/15)

    (a) Give the theorems for existence and uniqueness of a fixed point x* of a function g(x) and the convergence of the fixed point iteration. How do you use this method to solve the root of a function f(x)? Give an example.

    (b) Derive Newton's iteration starting from the fixed point iteration. Is this method always quadratic in convergence for any root x* of a function f(x)? If not, how do you modify the method to regain quadratic convergence given the nature of the root?

    (c) If f(x*)=0 and f'(x*) is not zero, then x* is also a solution to the function $g(x)=\frac{f(x)}{\sqrt{|f'(x)|}}$. Using this function in Newton's method, derive Halley's iteration given by:

$$x_{n+1}=x_n-\frac{2f(x_n)f'(x_n)}{2f'(x_n)^2-f(x_n)f''(x_n)}$$

2. SOLVE *ONE* ITERATION OF THE FOLLOWING: (/5 each)

    (a) Solution of
$$\begin{aligned}2x+3y-5z&=5\\ x^2+y^2+z^2&=2\\ x^3-yz&=1\end{aligned}$$
using Seidel's Fixed Point Iteration with $\begin{bmatrix}x_0\\y_0\\z_0\end{bmatrix}=\begin{bmatrix}0.5\\0.5\\0.5\end{bmatrix}$

    (b) Minimum of $z=(y-x^2)^2+(1-x)^2$ using Newton's Method with $\begin{bmatrix}x_0\\y_0\end{bmatrix}=\begin{bmatrix}2\\2\end{bmatrix}$

    (c) Nonlinear least squares solution for fitting $y=a\ln bx$ on the data set (2, 3.7), (4, 5.1), (7, 6.2) using Gauss-Newton's method with $\begin{bmatrix}a_0\\b_0\end{bmatrix}=\begin{bmatrix}1\\1\end{bmatrix}$

3. APPROXIMATION USING NEWTON'S METHOD (/10)

    (a) Using Newton's root finding method, derive an iterative method that computes for a real n^{th} root of a given number z using only the four basic arithmetic operations, namely: addition, subtraction, multiplication, and division.

    (b) Starting with x_0=1.5, find the (i) square root and (ii) cube root of 2. Since the method has quadratic convergence, the answer will be found after a few iterates only.

4. CONCEPTS (/5 each)

    1. If the bisection method starts with an interval whose length is one, i.e. $|a-b|=1$ and stops using terminal condition $|a-b|<tol$. After how many

iterations will the root in the interval be found?

2. What are the similarities and the differences between Regula-Falsi and Secant methods?
3. Illustrate graphical convergence of steepest descent and Nelder-Mead on a contour plot. (Note: gradient is perpendicular to the isoclines.)